

ADARSH CHARUGUNDLA

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EXPERIENCE

DevOps Engineer (Intern) | SkyIT (GBCS Group)

Dec 2025 – Apr 2026

Terraform, AWS (VPC, RDS, ECR), Kubernetes, Docker, OIDC, CI/CD

skyit.services

- Provisioned secure AWS infrastructure across three isolated environments (LokoMotive, Aukai, Orion) using Terraform IaC, designing custom VPC topologies with CIDR-level network segmentation
- Automated RDS deployments via reusable Terraform modules, authoring tfvars configurations and smoke-testing procedures that standardized provisioning, ensured high-availability storage, and reduced manual configuration errors
- Containerized frontend and backend services with Docker, building, tagging, and managing images through secure ECR pipelines for consistent deployments across all environments
- Built GitHub Actions CI/CD pipelines to automate build, test, and release stages, reducing manual intervention and ensuring stable, repeatable releases across all environments
- Orchestrated containerized workloads on Kubernetes across development and production, deploying via Docker images and ECR pipelines while managing pod scheduling, autoscaling, and service networking for reliable releases

PROJECTS

FrazyCorp IAM Least-Privilege Enforcement

github.com/AdarshChar/iam-least-privilege-audit

Terraform, AWS IAM, AWS Config, IAM Access Analyzer, GitHub Actions, OIDC, Python, boto3, Checkov

- Built a Terraform-managed IAM enforcement system for a simulated multi-department AWS environment, scoping custom least-privilege policies to specific resource ARNs across 4 departments and reducing overpermissioned roles from 4 to 0
- Automated continuous compliance auditing with AWS Config (5 rules) and IAM Access Analyzer, detecting real security gaps including missing MFA and overpermissioned policies with zero manual intervention
- Engineered a GitHub Actions CI/CD security gate using OIDC authentication to block pull requests containing wildcard or over-permissive managed policies before they reach AWS, eliminating stored-credential risk
- Developed Python automation scripts using boto3 to generate weekly IAM security reports covering MFA status, access-key age, policy violations, and Config compliance scores, saved automatically to S3 on a cron schedule

Traffic Sign Detection System

github.com/AdarshChar/GTSDB-Roadsign-Classifer

Python, YOLOv8, OpenCV, Roboflow, NumPy

- Trained YOLOv8n on the GTSDB dataset (500 road-scene images, 43 classes) achieving mAP@0.5 of 0.61 baseline and 0.65 post-augmentation at 12ms inference; documented findings in a full technical report covering per-class analysis across all 43 categories
- Engineered a custom preprocessing pipeline to convert PPM annotations to YOLO format, filtering unannotated images and resolving bounding-box coordinate corruption across 852 ground-truth entries
- Designed a photometric augmentation strategy (Gaussian blur, $\pm 20\%$ brightness, noise factor 1.13) to simulate adverse driving conditions, improving recall by 31.3% (0.501 $\rightarrow$ 0.658) with no architectural changes

TECHNICAL STRENGTHS

Languages

Python, JavaScript (ES6+), TypeScript, C/C++, SQL, Bash, HTML/CSS

Frameworks & Libraries

React, Node.js, Express.js, boto3

Cloud & DevOps

AWS, Terraform, Kubernetes, Docker, ECR, CI/CD, OIDC

Security & Compliance

IAM Least Privilege, AWS Config, Access Analyzer, Checkov, Static Analysis

AI & Data

YOLOv8, OpenCV, Roboflow, NumPy

Databases & Tools

PostgreSQL, MySQL, Microsoft SQL Server, Oracle, Git, Vim, Jest, JUnit

EDUCATION

University of Colorado Boulder

May 2026

B.A. in Computer Science & Engineering

Minor in Business

Cumulative GPA: 3.6